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SURVEY ON PREDICTION AND TASK AWARE SYSTEM FOR RESOURCE MONITORING IN VIRTUAL CLOUD ENVIRONMENT

Ananthi P
Assistant Professor, Rathinam Technical Campus, Coimbatore, Tamil Nadu, India.

ABSTRACT

Huge-rule statistics hubs control virtualization equipment towards reach tremendous source application, scalability, and high accessibility. Ideally, the act of an presentation successively exclusive a Virtual Machine (VM) shall be self-governing of co- located submissions and VM's that share the physical machine. However, adverse interfering effects exist and are especially severe for statistics- exhaustive applications in such virtualized environments. In this work, we present TRACON, a novel Task and Resource Allocation control framework that moderates the intrusion possessions from coexisting statistics-exhaustive applications. TRACON exploits modelling and control techniques from algebraic mechanism is demand consists of three major components: the interference prediction model that infers application performance from resource consumption observed from different VMs, the interference-aware scheduler that is designed to utilize the model for effective source management, and the task and resource monitor that collects application characteristics at the runtime for model adaption. The evaluation results show that TRACON can achieve up to 25 percent upgrading on application throughput on virtualized servers.

Key words: Cloud computing, virtualization, scheduling.

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1. INTRODUCTION

Cloud computing has achieved tremendous success in offering Infrastructure / Platform / Software as a Service, in an on-demand fashion, to a large number of clients. This is evident in the popularity of cloud software services, e.g., Gmail and Face book, and the rapid development of cloud platforms, e.g., Amazon Elastic Compute Cloud (EC2). The key enabling factor for cloud computing is the virtualization technology, e.g., Xen , that provides an abstraction layer on top of the underlying physical resources and allows multiple operating systems and applications to simultaneously run on the same hardware. As virtual machine monitors (VMM) encapsulate different applications into each separate guest virtual machine (VM), a cloud provider can leverage VM consolidation and migration to achieve excellent resource utilization and high availability in large data centers. Ideally, an application running inside a VM shall achieve the performance as it would own a portion of the machine to itself, that is, independent of co-located applications and VMs that share the same physical resource. Although extensive work has been done to achieve this so-called performance isolation including various techniques to ensure CPU fair sharing. Little attention has been paid to data-intensive applications that perform complex analytics tasks on a large amount of data, which have become increasingly common in this environment. Traditionally assuming the exclusive ownership of the physical resources, these applications are optimized for the hard drive based storage systems by issuing large sequential reads and writes. To the contrary, multiple data-intensive applications will be in competition for the limited bandwidth and throughput to network and storage systems which very likely leads to high I/O interference and low performance. In this case, the combined effects from concurrent applications, when deployed on shared resources, are largely difficult to predict, and the interference as a result of competing I/Os remains problematic to achieve high-performance computing in a virtualized environment. In this work, we study the performance effects of collocated data-intensive applications, and develop TRACON, a novel Task and Resource Allocation Control framework that mitigates the interference from concurrent applications. TRACON leverages modeling and control techniques from statistical machine learning and acts as the core management scheme for a virtualized environment.

The experiments on real-world cloud applications show that TRACON can achieve up to 25 percent improvement on application throughput for virtualized data centers. We characterize the I/O interference from multiple concurrent data-intensive applications in a virtualized environment, and build interference prediction models that reason about application performance in the event of varied levels of I/O interference. While prior work has extensively studied the VM interference on CPUs, caches, and main memory, we address the new challenges that arise when modeling data-intensive applications in virtualized data centers. Our statistical models profile the performance of a target application, when running against a set of benchmarks, to infer the runtime of the application. We propose to use nonlinear models that are critical to capture the burstyI/O patterns in data-intensive applications. This design utilizes the application characteristics, observed from the VMs, and maintains a low system overhead. We develop TRACON for virtualized data centers to mitigate the interference effects from co-located data-intensive applications. To achieve this goal, we incorporate the proposed nonlinear interference prediction models into TRACON and by doing so, the system can make optimized scheduling decisions that lead to significant improvements in both application performance and resource utilization. We conduct a comprehensive set of experiments on two virtualized clusters with real-world cloud applications.

2. BACKGROUND

Virtualized data centers are common cloud computing platforms. In this work, we focus on Xen and its notable para virtualization

Technique, where Xen VMM works as a hardware abstraction layer to guest operating systems with modified kernels. In para virtualization, the VMM is in Charge of resource control and management, including CPU time scheduling, routing hardware interrupt events, allocating memory space, etc. In addition, a driver domain (Dom0) that has the native drivers of host hardware performs the I/Operations on behalf of guest domains (DomU).When multiple VMs are running on the same physical machine, several factors contribute to the degraded application performance, including virtualization overheads and the imperfect performance isolation between VMs.

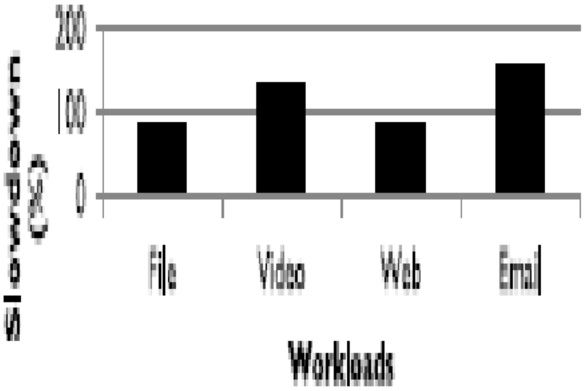

Figure 1. Slowdowns by competing VMs in Amazon EC2.

We have illustrated the interference problem with experiments on local machines. A similar test can also be demonstrated in a public cloud system. The competition between virtual I/O workloads, an adversary can drag down the performance of a VM that shares the same resources. This test is conducted on Amazon EC2 with the small instance. After we locate VM1 and VM2 that share the same I/O resources, VM2 competes for the I/O bandwidth when it detects that VM1 is heavily consuming I/O resources. I/O interference is crucial for virtualized data centers to improve customer experiences and increase resource utilization. In this work, we build interference prediction models that automatically infer the effects when there exist varied levels of concurrent I/O operations, and propose novel scheduling algorithms to minimize the negative impacts of co-located applications.

3. TRACON SYSTEM ARCHITECTURE

Most cloud service providers utilize a hierarchical management scheme to administer the large quantity of machines. In this environment, each application server has the same virtualized environment, where VMs are dynamically allocated to run applications from the clients. A manager server is responsible for supervising a group of application

Servers, which report their status in a time interval to the manager server. The manager servers can form a tree-like hierarchy for high scalability. Given a virtualized environment that consists of a large number of physical machines and different applications, we utilize statistical machine learning techniques, in particular statistical modeling for reasoning about the Application's performance under interference.

We share the same philosophy that the statistical machine learning will play an important role in the application and resource management in large-scale data centers. Upon the arrival of tasks, the scheduler will generate a number of possible assignments based on the incoming tasks and the list of available VMs, which will then be communicated to the interference prediction module. Assigns the tasks to different servers.

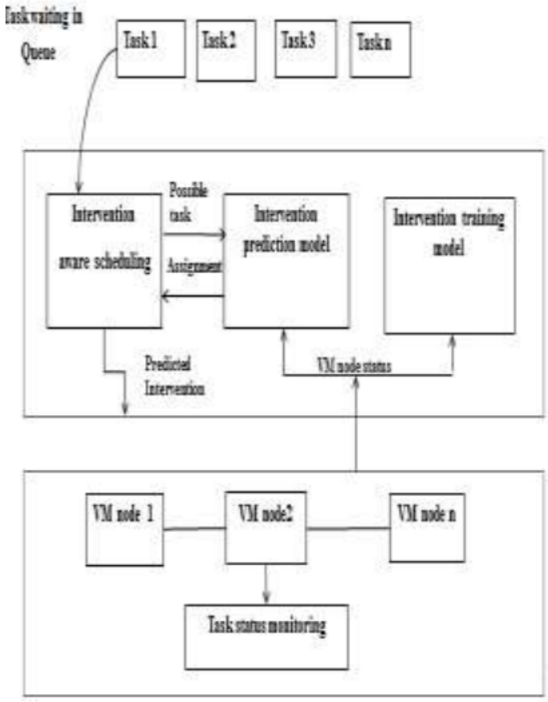

Figure 2. Tracon Architecture

3.1. Interference Prediction Model

On the high level, the interference can be perceived as the changes in the performance, including the total runtime as Used in prior work and I/O throughput that we have shown is critical to data-intensive applications. In this work, we construct the interference prediction models in order to extrapolate the application performance as a function of the resource consumption by virtual machines.

3.2. Interference-Aware Scheduling

In general, optimally mapping tasks to machine in parallel and distributed computing environments has been shown to be an NP-complete problem. In this work, we explore a number of heuristic techniques to find a good solution for the scheduling problem. Specifically, TRACON aims to reduce the runtime and improve the I/O throughput for data-intensive applications in a virtualized environment.

K-means++:

The quality of final clustering of a k-means algorithm mainly depends on the initialization process. The idea of most k-means algorithms is to pick points that are far away from other.

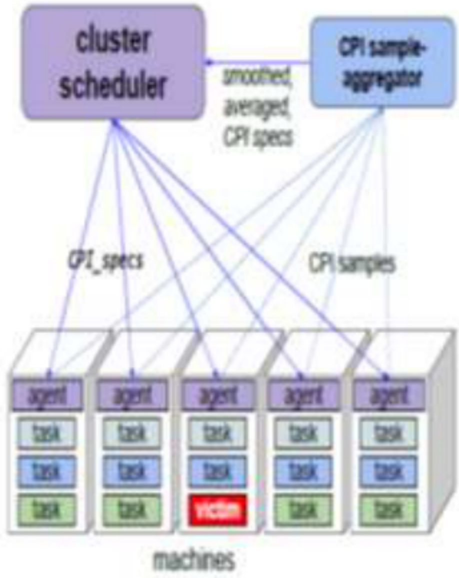

Figure 3. The CPI2 data pipeline.

VM performance modeling, measure and use application characteristics to model the virtualization overheads. VM profiling for performance. Debugging and performance bottleneck in a virtualized environment. The CPI data is sampled periodically by a system daemon using the perf event tool in counting mode

To keep overhead to a minimum. We gather data for a 10 second period once a minute; we picked this fraction to give other measurement tools time to use the counters. Google-Wide Profiling (GWP) [31] gathers performance counter Sampled profiles of both software and hardware performance events on Google's machines. It is active on a tiny fraction of machines at any time, due to concerns about the overhead of profiling. In contrast, CPI2 uses hardware performance

Counters in counting mode, rather than sampling, which lowers the cost of profiling enough that it can be enabled on every shared production machine at Google at all times.

MIMO power controller that adjusts power among servers based on their performance needs, while controlling the total power of the cluster to stay at or below a constraint imposed by the capacity of its power supplies. A two-level control system to manage the mappings of workloads to VMs and VMs to physical resources. It focuses on a multi-objective optimization problem aiming to simultaneously optimize possibly conflicting objectives, including making efficient usage of multidimensional resources, avoiding hotspots and reducing energy consumption.

PSciMapper is power-aware consolidation framework for scientific workloads and builds the models to relate the resource requirements to performance and power consumption. PSciMapper evaluates the tradeoff between

Energy reduction and performance degradation when consolidating workloads onto one host, and utilizes correlation analysis between performance and resource contention as the distance metric in its clustering algorithm for consolidation. There is no direct performance prediction in pSciMapper.

4. CONCLUSION

In this work, we investigate the performance effects of co-located data-intensive applications in virtualized environments, and propose a management

System TRACON that mitigates the interference effects from concurrent data-intensive applications and greatly improves the overall system performance. In future work, we will be exploring adaptive throttling and making job place improves the overall system performance. First, we study the use of statistical modeling techniques to build different models of performance interference, and propose to use the non-linear models as the prediction module in TRACON. Second, we develop several scheduling algorithms that work with the prediction module to manage the task assignments in virtualized data centers. We also integrate VM migration and consolidation in the management system. TRACON achieves up to 25 percent improvement on throughput for real-world cloud applications.

REFERENCE

[1] F. Nadeem and T. Fahringer, "Predicting the Execution Time of Grid Workflow Applications through Local Learning,"Proc.Conf.High Performance Computing Networking, Storage and Analysis (SC'09), 2009.

[2] REN, G., TUNE, E., MOSELEY, T., SHI, Y., RUS, S., AND HUNDT, R. Google-Wide Profiling: a continuous profiling infrastructure for data centers. IEEE Micro, 4 (July 2010), 65–79.SANCHEZ.

[3] X. Wang and M. Chen, "Cluster-Level Feedback Power Control for Performance Optimization,"Proc. IEEE 14th Int'l Symp. High Performance Computer Architecture (HPCA '08), pp. 101-110, Feb.2008

[4] Q. Zhu, J. Zhu, and G. Agrawal, "Power-Aware Consolidation of Scientific Workflows in Virtualized Environments," Proc. ACM/ IEEE Int'l Conf. for High Performance Computing, Networking, Storage and Analysis (SC '10), 2010.

[5] R.C. Chiang and H.H. Huang, "Tracon: Interference-Aware Scheduling for Data-Intensive Applications in Virtualized Environments,"Proc. Int'l Conf. for High Performance Computing, Networking, Storage and Analysis, pp. 47:1-47:12, 2011.